

## **SAFETY QUIZ:CHE382-Chemical Engineering Laboratory II**

Please answer the following questions using the information from the 6 selected chapters of the JT Baker Hazardous Chemical Safety document; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards and Compressed Gases given in Safety Assignment I. Note: Answers from another source may not be considered correct. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words. (5 points per question)

1. What is the definition of a toxic chemical?

A Toxic Chemical is defined as a chemical that when ingested, inhaled or absorbed or when applied to, injected into or developed within the body, in small amounts by its chemical action that can cause damage to structure or disturbance to function according to Dorland's medical dictionary.

2. Name the 4 types of toxicity damage from a toxic chemical:

The 4 types of toxicity damage from a toxic chemical are the following Local Toxicity, systemic toxicity, Acute Toxicity, and Chronic Toxicity.

3. Name two parameters that can affect toxicity:

The two parameter that may affect toxicity is the physical condition of the individual affected by the chemical and the mode of entry of the toxic chemical.

4. What is the PEL (Permissible Exposure Limit)?

Permissible Exposure Limit is defined as the maximum exposure recommended by NIOSH which is commonly derives from the TWA values given by ACGIH and are enforced by OSHA.

5. What is the DOT (Department of Transportation) definition of a compressed gas?

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The DOT definition of a compressed gas is the following Compressed gases are defined as those gases that have a psia greater the 40 at 70 degrees Fahrenheit or a psia greater 104 at 130 degrees Fahrenheit or those flammable materials that have a greater psia than 40 at 100 degrees Fahrenheit

6. a) How does one determine how much material remains in a non-liquified gas cylinder?

The method as to how to determine how much material remains in a non-liquefied gas cylinder is the by the pressure gauge reading as a non-liquified gas cylinder is entirely gas so therefore the amount of gas within the container is directly proportional to the pressure gauge reading.

- b) On page 13-10, which pressure gauge on the two-stage regulator will give you the cylinder pressure reading?

The pressure gauge that will provide you with the cylinder pressure reading is the gauge furthest to the left showing the cylinders pressure.

7. How does one determine how much material remains in a liquified gas cylinder?

The method of determining how much material remains in a liquified gas cylinder is by weighing it as in this case the cylinder has liquids in it therefore the pressure remains are constantly the same therefore you have to use weight instead.

8. Why do insidious hazards represent a problem to workers?

Insidious hazards represent a problem to workers because they may never know that these hazards are present until they have experienced chronic problems.

9. What are the four general categories of insidious hazards?

The four general categories of insidious hazards are the following Improper handling of equipment or chemicals, unexpected chemical reactions, Unanticipated Toxicity, and Improper Faculty design.

10. Name 2 control measures that can be implemented to reduce insidious hazards in the laboratory.

Two control measures that can be implemented to reduce insidious hazards in the laboratory are the following first is to have adequate ventilation on top of having proper fume hoods in order to handle chemicals and having the proper PPE with proper hygiene.

11. What is the definition of a reactive chemical?

A reactive chemical is a chemical with the capabilities of going under rapid chemical change under normal or slightly different conditions where the reaction can release heat, generate gas, cause an explosion, fire or any other toxic byproducts.

12. Name the two groups and their sub-headings used to categorize reactive chemicals:

The two groups used are Unstable chemicals and unstable mixtures.

Unstable Chemicals- subheadings ( explosive, self reactive, and heat sensitive chemicals)  
Unstable Mixtures – subheadings (Water reactive chemicals, Air Reactive, and Oxidizers)

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13. List two precautionary measures for handling and/or storing reactive chemicals.

The first precautionary measure is making sure to store chemicals away from incompatible materials and the second is to make to use proper containers where reactive chemicals should be in approved containers as well as that prevent anything from entering.

14. How are corrosive chemicals generally defined?

Corrosive chemicals are defined as chemicals that cause visible destruction or irreversible damage to the skin or other body tissue at the site of contact.

15. How do corrosive chemicals act on body tissues?

Corrosive chemicals act on the skin through direct contact causing pain, redness, and burns causing severe tissue damage.

16. True or False: Concentrated alkaline solutions have a more corrosive effect on tissue than most strong inorganic acids.

True

17. Name the six categories of common corrosive liquids.

The six categories of common corrosive liquids Inorganic Acids, Organic Acids, Caustic Solutions, Organic Solvents and other Organics.

18. What category of corrosive solids presents the greatest hazard because of their frequent use in industry?

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The category of corrosive solids that presents the greatest hazard because of their frequent use in industry is caustic alkalizes.

19. Name the three components that make up the FIRE TRIANGLE:

The three components that make up the Fire triangle is Fuel, Oxidizer, and energy.

20. What category of flammables is the most common source of fires in industry?

Flammable Solvents are the most common source of fires in industry.